A swarm that owns a proof can still lose it

Simulation 1, setup `task003-symmetric`: first results

LLM-free simulator; extended with the efficiency of control, 12 October 2026; runs of 7 October 2026

How the simulator works. The world, `task_003`, has three candidate allocations; A0 is the correct one. Each of 24 agents is given one true fact about it and reasons as an exact Bayesian: its *belief* is the exact probability of A0, A1 and A2 given the facts it currently remembers. An episode has 30 rounds, each a day and a night. During the day the agents act one at a time in a new random order: an agent reads q posts drawn at random from a shared *board* (never its own), adds their facts to its memory, recomputes its belief, *votes* (*argmax*: the most probable allocation; *probability matching*: a vote drawn from its belief) and posts one remembered fact that supports its vote, or nothing if none does. At night a *controller* with a target, A0 (truth) or A2 (false), reads q_c agent posts; unless at least 75% of them come from agents voting for its target, or what it read already proves its target, it posts the b facts from its own pool that best support its target. The board is then cleared, keeping only the controller's new posts. At each dawn every remembered fact is forgotten with probability $1 - \rho$ and comes back if read again. Every episode is run three times with the same random draws: without a controller, with the truth controller and with the false controller.

Summary. 24 agents together hold a complete proof that allocation A0 is correct, but no single agent does. A controller steers toward A0 (truth) or A2 (false). (i) **Whether the false controller wins depends on memory**: when agents forget, it overturns the proof; with full memory its success is temporary, because its own true facts end up completing proofs of A0. (ii) **The controller senses the swarm accurately**, so its failures are not sensing failures. (iii) **Helping the truth is not the same as helping agents assemble their own proof**: extra truth posts raise the share of agents holding a proof, but lower the share holding one from their own evidence.

1 Setup

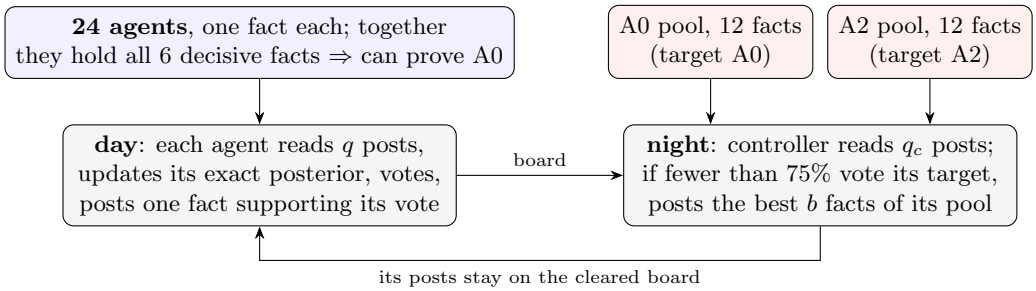


Diagram 1. One round of Simulation 1: a day followed by a night, repeated for 30 rounds. Each dawn, every remembered fact survives with probability ρ.

The two pools share no fact, contain no decisive fact, cannot prove their target, and are matched in strength pair by pair. The agents start, in expectation, with 9 / 7 / 8 votes for A0 / A1 / A2. Agents are exact Bayesian reasoners: an agent's belief is the exact posterior over the three allocations given the facts it remembers. Three agent rules are compared: *argmax* voting with posts that support the vote (the *base* rule), *probability matching* (vote drawn from the posterior), and argmax with *uniform* posts (a random remembered fact, whatever the vote).

Measuring control. Every episode is run three times with the same random draws: silent, truth controller, false controller. The *paired gain* of a controller in the share of agents voting for allocation a on day t is

$$G_t^{(a)} = \frac{1}{E} \sum_{e=1}^{E} \Big[x_{a,t}^{\text{ctrl}}(e) - x_{a,t}^{\text{silent}}(e) \Big], \tag{1}$$

with $E = 1000$ episodes per cell and 95% intervals from resampling episodes.

2 A swarm that owns a proof can still lose it

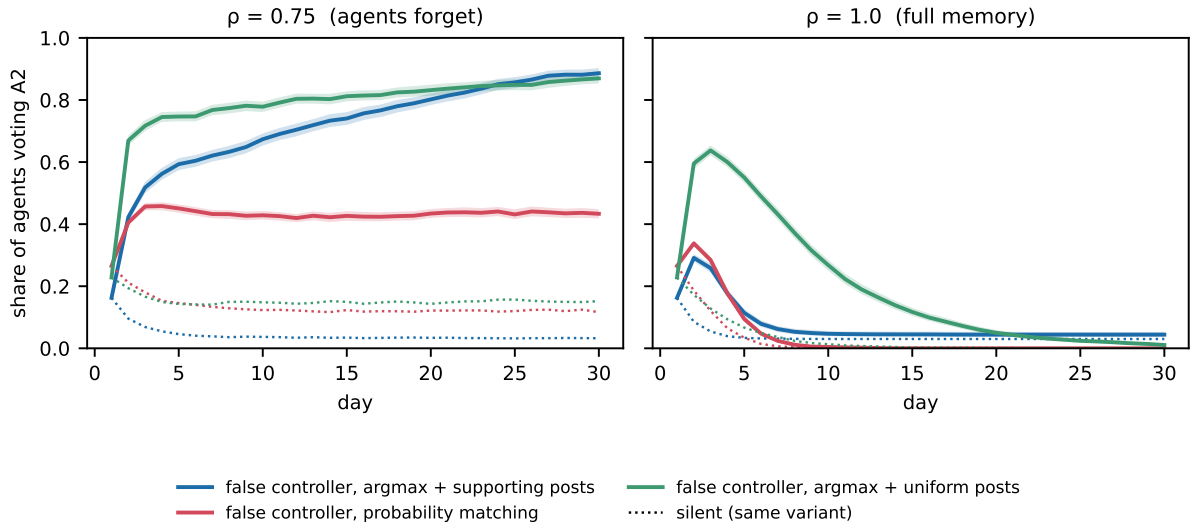


Figure 1: Share of agents voting A2 over the days, under the false controller (solid, 95% band) and without a controller (dotted). $q = 6$, $q_c = 12$, $b = 3$.

Figure 1 shows the central result. **When agents forget** ($\rho = 0.75$), the false controller pushes the A2 share from under 0.2 to about 0.9 under the base and uniform rules, and to about 0.45 under probability matching. **With full memory** ($\rho = 1$) every rule shows the same shape: an early A2 peak (up to 0.6 under uniform posts), then a return to A0 by day 20. Table 1 gives the day-30 gains.

Table 1: Day-30 paired gain of the false controller in the A2 share, Eq. (1); $q_c = 12$, $b = 3$.

ρ	agent rule	$q = 3$	$q = 6$	$q = 12$
0.75	argmax + supporting posts	0.95	0.85	0.26
0.75	probability matching	0.48	0.32	0.08
0.75	argmax + uniform posts	0.67	0.72	0.69
1.0	argmax + supporting posts	0.03	0.01	0.00
1.0	probability matching	0.00	0.00	0.00
1.0	argmax + uniform posts	0.05	0.01	0.01

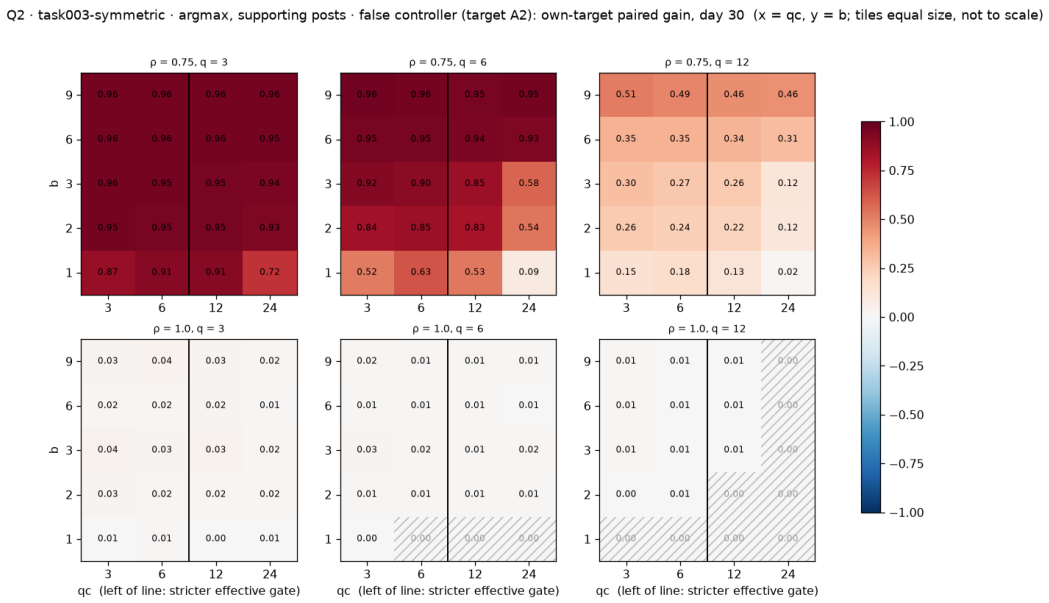


Figure 2: Day-30 paired gain of the false controller (A2 share) over the reading budget q_c (x) and posting budget b (y); base rule. Rows: ρ; columns: q. Tiles are equal size, not to scale. Left of the black line, the vote gate is effectively stricter (3 of 3, 5 of 6 posts).

The phase diagram (Figure 2) separates the three factors that matter:

- **Memory ρ decides** whether the false controller can win at all (top row against bottom row).
- **Reading more defends the proof** under supporting posts: at $\rho = 0.75$ the gain falls from about 0.9 at $q = 3$ to about 0.3 at $q = 12$. Agents who hold pieces of the proof keep posting them, so readers keep meeting them. Under uniform posts this defence disappears (Table 1: 0.67–0.72 at every q).
- **A larger posting budget helps the false controller** ($b = 1 \to 9$: 0.52 → 0.79 at $\rho = 0.75$, averaged over q): each pool is entirely on its target's side, so more posts mean more evidence for A2. The reading budget q_c matters little.

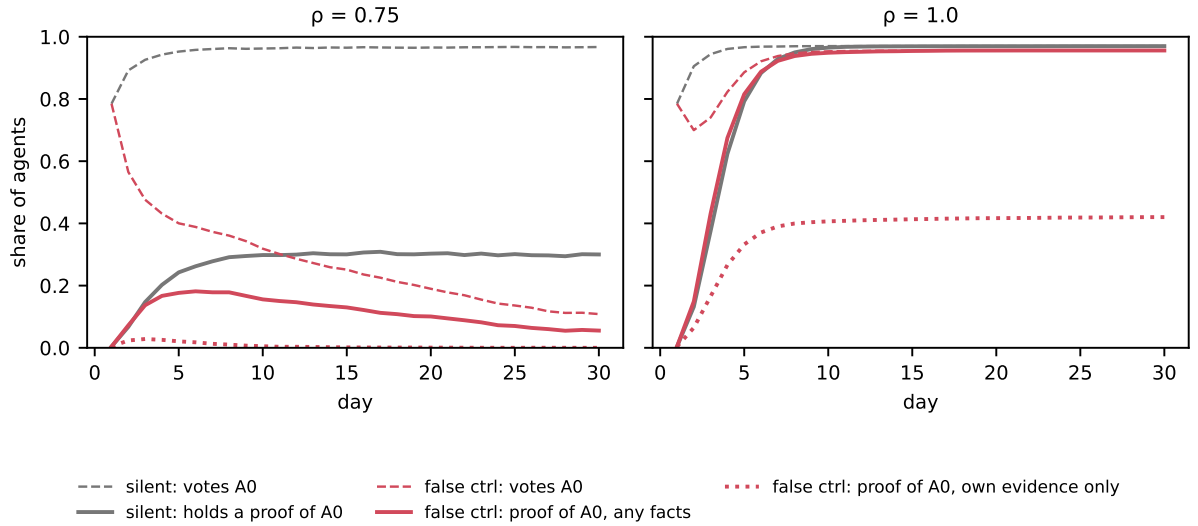


Figure 3: Share of agents voting A0 (dashed) and holding a proof of A0 (solid), silent against false controller. Dotted: proof from the agents' own original evidence only. Base rule, $q = 6$, $q_c = 12$, $b = 3$.

Why memory decides. Figure 3 shows the mechanism. With forgetting, a proof must be reassembled continually; while the false controller fills the board with A2 evidence, the agents' proof decays (own-evidence proofs fall to zero) and the A0 vote follows. With full memory the opposite happens: agents accumulate facts, including the controller's A2 facts. **Those facts are true**, so together with the agents' evidence they complete

proofs of A0: on day 30, 95.6% of agents hold a proof of A0, but only 42% from their own evidence. The rest are proofs the false controller helped build. Every fact a truthful controller posts eventually narrows the possible worlds toward the real one, and with full memory nothing is lost.

3 The controller senses the swarm accurately

Each night the controller estimates the share of agents backing its target from the $n \le q_c$ agent posts it reads:

$$\hat{v} = \frac{1}{n} \sum_{i=1}^{n} \mathbb{1}\{\text{author of post } i \text{ voted for the target}\}, \tag{2}$$

and stays silent when $\hat{v} \ge 0.75$. Each agent posts at most once a day, so the day's board is nearly a census of votes and $\hat{v}$ is an unbiased sample of it.

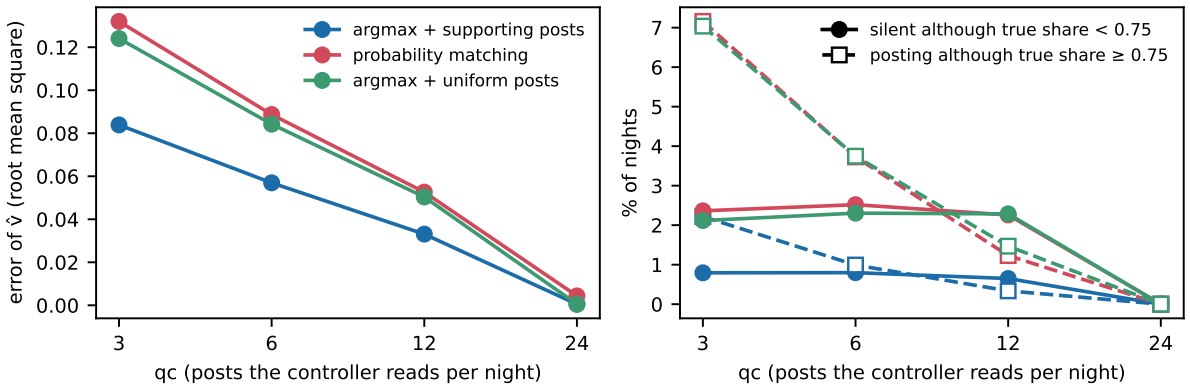


Figure 4: Left: error of $\hat{v}$ against the true share of the 24 agents. Right: gate mistakes. Mean over arms, q, b, ρ; task003-symmetric.

- **No bias**: $\hat{v}$ minus the true share averages within ± 0.0005.
- **The error shrinks with** q_c as random sampling predicts, and vanishes at $q_c = 24$, where the controller reads every post (Figure 4, left).
- **Gate mistakes are rare** (right panel): under 2% of nights at $q_c \ge 12$. They grow at $q_c = 3$, where the gate needs all 3 posts to agree.
- **Abstention does not distort it**: agents almost never abstain (0.1% of agent-days, probability matching only).

So in this setup, **a controller fails because of what it posts, not what it sees.**

4 Helping the truth is not helping agents assemble their own proof

The truth controller stays silent when its sample of the board already proves A0 (`silent_when_target_proved`). We reran all its cells with that rule off and compared, episode by episode, two measures: an agent holds a proof of A0 (i) from any facts it remembers; (ii) from its own original evidence only.

Table 2: Rule off minus rule on, day 30, truth controller, mean over q, q_c, b. "Rule fires": share of nights the controller sees A0 proved; "posts despite proof": share of nights it posts although A0 is proved.

agent rule, ρ	rule fires (on)	posts despite proof (off)	Δ proof, any facts	Δ proof, own evidence
base, 0.75	29%	0.3%	+0.006	−0.012
base, 1.0	32%	0.3%	0.000	−0.029
prob. matching, 0.75	34%	2.3%	+0.016	−0.044
prob. matching, 1.0	31%	0.8%	0.000	−0.057

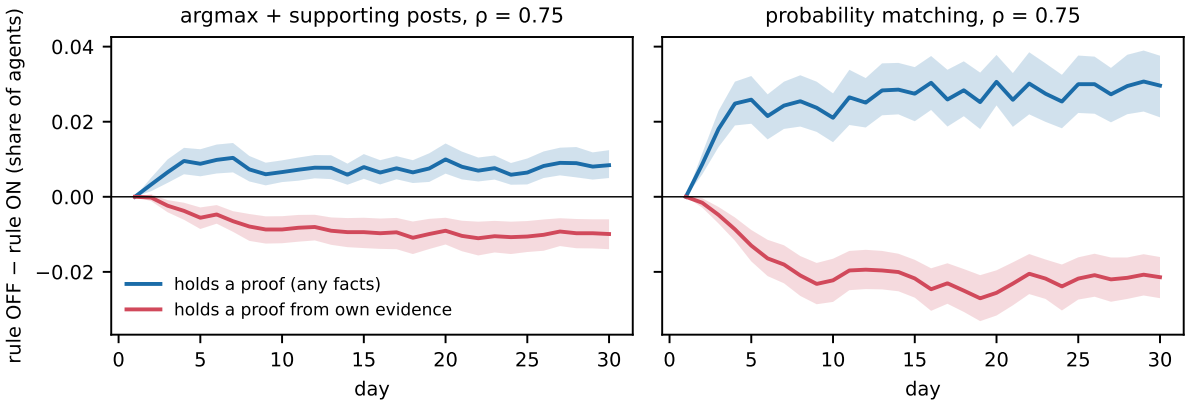


Figure 5: Rule off minus rule on, over days: proof from any facts (blue) and from own evidence (red). $q = 6$, $q_c = 12$, $b = 1$, $\rho = 0.75$.

- **The vote gate already does most of the stopping**: the rule fires on about 30% of nights, yet with it off the controller posts on at most 2% of nights, because by then 75% of the posts it reads already vote A0 (Table 2).
- **The few extra posts pull the two measures apart** (Figure 5): more agents hold a proof, but fewer hold one from their own evidence.
- **Mechanism: posting, not reading.** Controller posts are at most 1.5% of what agents read. But an agent passes on one fact a day, and a controller fact it read on an earlier day now competes with its original evidence for that slot (Table 3). The original evidence circulates less, so fewer agents assemble it into a proof.
- Figure 3 shows the same split for the false controller, much larger: at $\rho = 1$, 95.6% of agents hold a proof of A0 but only 42% from their own evidence.

Table 3: What agents read and post, rule on vs off. Truth controller, probability matching, $\rho = 0.75$, $q = 6$, $q_c = 12$, $b = 3$, all days.

rule	reads from controller	posts of own evidence	posts of pool facts	own facts held
on	1.34%	78.3%	27.3%	5.81
off	1.50%	76.1%	29.3%	5.67

5 How efficiently does the controller steer?

A controller can change the swarm in two ways: in the direction it is asked to, or in ways that happen whatever its target. We measure how readable the command is in the swarm, how far each controller gets toward its target, how much of its change is directed, what each fact it posts buys, and how unsure the agents become. Throughout, the target is chosen by a fair coin, and all quantities are taken at one time t, across the 1,000 episodes of each arm (not over whole trajectories). Setting: probability matching, $q = 6$.

5.1 Measures

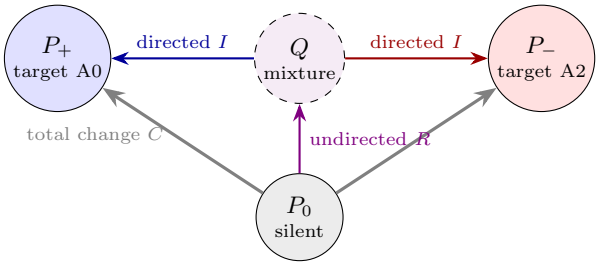


Diagram 2. Three distributions of the swarm's state at time t, one per arm, each over the episodes of that arm. The controller's total change C splits exactly into a part that depends on its target (I, from the mixture Q to each arm) and a part that does not (R, from silence to Q). Efficiency $\eta = I/C$ is the directed share.

Distributions. For each arm a (silent 0, target A0 $+$, target A2 $-$) and a time t, each episode e contributes a probability vector $y_a(e)$ over the values x of the state X: at *one-agent* level its vote shares (the vote of a randomly chosen agent), at *swarm* level a mark on the bin of $s = m_{\mathrm{A0}} - m_{\mathrm{A2}}$ (difference of mean beliefs, three equal bins on $[-1, 1]$). Then $\hat{P}_a = \frac{1}{n}\sum_e y_a(e)$. Any pair of controllers can be written as

$$\hat{P}_\pm = \hat{P}_0 + c \pm \Delta, \qquad c = \tfrac{1}{2}(\hat{P}_+ + \hat{P}_-) - \hat{P}_0, \quad \Delta = \tfrac{1}{2}(\hat{P}_+ - \hat{P}_-), \tag{3}$$

a *common* part c (what control does on average, whatever the target) and a *directional* part Δ.
Readability and the pair efficiency. With $\hat{Q} = \hat{P}_0 + c$ and $\mathrm{KL}(p\|q) = \sum_x p(x)\log_2[p(x)/q(x)]$,

$$I = \tfrac{1}{2}\mathrm{KL}(\hat{P}_+\|\hat{Q}) + \tfrac{1}{2}\mathrm{KL}(\hat{P}_-\|\hat{Q}), \quad C = \tfrac{1}{2}\mathrm{KL}(\hat{P}_+\|\hat{P}_0) + \tfrac{1}{2}\mathrm{KL}(\hat{P}_-\|\hat{P}_0) = I + R, \quad R = \mathrm{KL}(\hat{Q}\|\hat{P}_0), \quad \eta = I/C. \tag{4}$$

I is the mutual information between the target and the state: how readable the command is (at most 1 bit). I depends only on Δ, R only on c; the reference $\hat{P}_0$ is the silent arm. For small changes, $\mathrm{KL}(p + \epsilon\|p) \approx \sum_x \epsilon(x)^2/(2\ln 2\, p(x))$ gives

$$\eta \approx \frac{1}{1 + \|c\|^2/\|\Delta\|^2}, \qquad \|v\|^2 = \sum_x \frac{v(x)^2}{\hat{P}_0(x)}. \tag{5}$$

So $\eta = 1$ exactly when the two controllers' changes are mirror images ($c = 0$), whatever their size, and η falls only through the *ratio* of common to directional change: when both push the same way (a common push), or one moves the swarm much more than the other (a lopsided pair: along one direction with sizes a and b, $\|c\|/\|\Delta\| = |a-b|/(a+b)$). It is a property of the pair, not of either controller, and it stays constant whenever both effects grow together. At swarm level C is often infinite (the controlled swarm reaches states silence never visits); there we use $\eta_{JS} = I/(I + \mathrm{JS}(\hat{Q}, \hat{P}_0))$ with the bounded Jensen–Shannon divergence.
Each controller against its own target z (one agent, votes):

$$S_z = \mathrm{KL}\big(\mathrm{Bern}(\hat{P}_z(z)) \,\|\, \mathrm{Bern}(\hat{P}_0(z))\big), \qquad \pi_z = \frac{S_z}{\log_2[1/\hat{P}_0(z)]}, \qquad \varphi_z = \frac{S_z}{\mathrm{KL}(\hat{P}_z\|\hat{P}_0)}, \qquad \text{price} = \frac{S_z}{\text{facts posted}}. \tag{6}$$

S_z is the least information any intervention must inject to raise the target's share from its silent value to the observed one (raise the target, scale the other answers in proportion). **Progress** π_z is the share of the way from silence to unanimity on the target, in bits; **focus** φ_z the share of the controller's actual change that was needed for that gain; both are negative when the controller moved its target *down*. The **price** is the information injected per fact the controller posted. Finally, the agents' own uncertainty: $\Delta\bar{H}$ (mean entropy of their beliefs) and ΔJSD (disagreement, the entropy of the mean belief minus $\bar{H}$), controlled minus silent, same episode.

5.2 Over time, at one setting

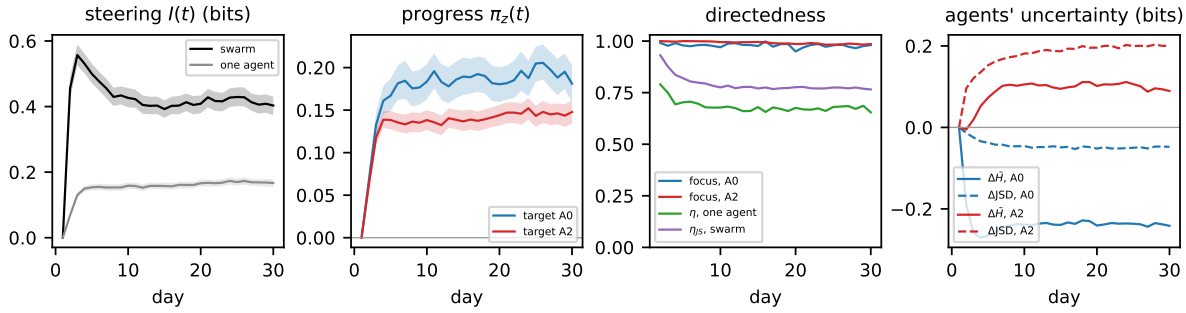


Figure 6: At $\rho = 0.75$, $q_c = 12$, $b = 3$ (end of each day): readability $I(t)$, each controller's progress $\pi_z(t)$, directedness (focus φ_z, pair efficiency η, η_{JS}), and the agents' uncertainty. Bands: 95% intervals over episodes.

- **Control is reached within a few days and then stays.** Both controllers get 90% of their day-30 progress by day 4–5: $\pi_{\mathrm{A0}} = 0.18$, $\pi_{\mathrm{A2}} = 0.15$ at day 30.
- **Both controllers are focused** ($\varphi \geq 0.97$): almost all of what each changes moves its own target. The pair efficiency is nevertheless only 0.65 (η_{JS} 0.77): the pair is lopsided (next subsections).
- **The false controller confuses**: it makes agents less sure ($\Delta\bar{H} = +0.09$ bits) and splits them (ΔJSD $= +0.20$); the truth controller makes them surer (-0.24) and more alike (-0.05).

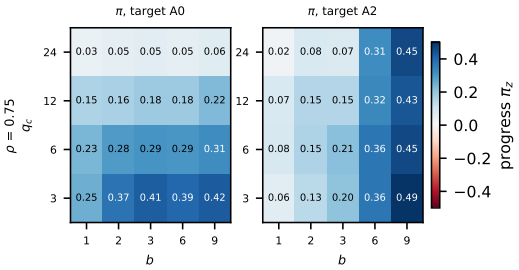


Figure 7: Progress π_z of each controller at $t = 30$ over the reading budget q_c and the facts per night b (one agent, votes). Red: the controller moved its target down. Dash: undefined (silence already at unanimity).



Figure 8: Left: focus φ_z (negative: the change went against the target). Right: price, information injected per fact posted (bits). $t = 30$; Dash: undefined (no change, or the controlled swarm considers an answer silence rules out completely).

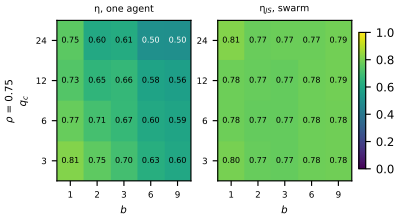


Figure 9: Pair efficiency at $t = 30$ over q_c and the facts per night b: one-agent η (beliefs) and swarm η_{JS}. Dash: undefined.

5.3 Across the control plane

- **The false controller's progress grows with** b, from 0.02–0.08 at $b = 1$ to 0.43–0.49 at $b = 9$; the truth controller's is set by q_c (0.25–0.42 at $q_c = 3$, 0.03–0.06 at $q_c = 24$): the silent swarm already sits at 85% A0, so its vote gate (at least 75% of the posts read) keeps it silent unless it reads few posts.
- **Focus stays at 0.93 or more everywhere**: neither controller wastes its change.
- **Each extra fact buys less**: the price (target A0) falls from up to 0.020 bits per fact at $b = 1$ to 0.002–0.005 at $b = 9$.
- **With full memory** every agent proves A0 and nothing is measurable.

5.4 Readability, pair efficiency and uncertainty against the facts per night

- η **falls with** b **(0.73 to 0.56) because the pair becomes lopsided**, not because either controller wastes effort: the false controller moves the A0 share from 85% to 27% at $b = 9$, the truth controller only to 93% (near the ceiling). On average control then shifts the swarm toward A2; that shift is the same whatever the target, so η counts it as undirected.
- Readability rises with b (swarm 0.24 to 0.94 bits).

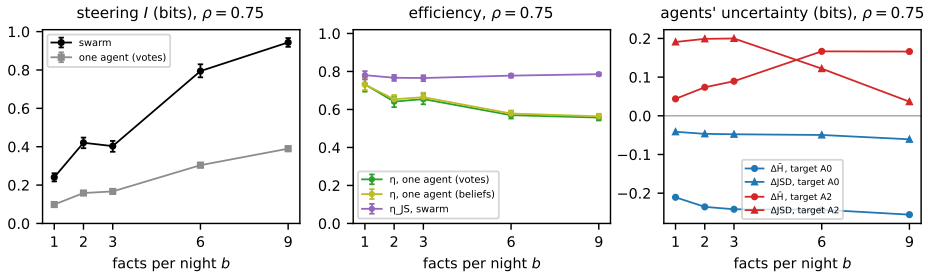


Figure 10: Readability I (left), pair efficiency (middle) and the change in the agents' uncertainty (right; circles $\Delta\bar{H}$, triangles ΔJSD) at $t = 30$ against facts per night b ($q_c = 12$). Bars: 95% intervals.

5.5 Precision at 50, 100 and 300 paths

Table 4: Value from 1,000 paths per arm, and root-mean-square error of the estimate from 50, 100 or 300 paths (200 random subsets each), at $t = 30$, $b = 3$, $q_c = 12$, $\rho = 0.75$. *: share of subsets in which the quantity is infinite or undefined.

quantity	1,000 paths	error, 50	error, 100	error, 300
steering I, swarm (3 bins)	0.403	0.059	0.042	0.023
steering I, one agent (votes)	0.167	0.023	0.016	0.009
efficiency η, one agent (votes)	0.655	0.063	0.044	0.022
efficiency η_{JS}, swarm	0.765	0.038	0.029	0.012
progress π, target A0 (votes)	0.181	0.049	0.034	0.015
progress π, target A2 (votes)	0.148	0.028	0.021	0.011
focus φ, target A0 (votes)	0.981	0.057	0.039	0.016
focus φ, target A2 (votes)	0.986	0.016	0.010	0.005
$\Delta\bar{H}$, target A0	-0.242	0.031	0.020	0.010
$\Delta\bar{H}$, target A2	0.089	0.035	0.024	0.012
ΔJSD, target A0	-0.048	0.011	0.006	0.004
ΔJSD, target A2	0.200	0.021	0.018	0.009

Across all settings, at 100 paths: readability ± 0.01 (one agent) to ± 0.03 (swarm), progress ± 0.02, focus ± 0.01–0.02, η ± 0.02–0.03, $\Delta\bar{H}$ ± 0.02, ΔJSD ± 0.006. Two settings that differ by at least 0.02 are ordered correctly in 90–100% of subsets.

Caveats. One world (task_003), 24 agents, exact-Bayesian agents; numbers are means over 1,000 episodes per cell. Full analysis, code and every table: `analysis/task003_llm_free/sim1_analysis/` (reports Q0–Q6).